

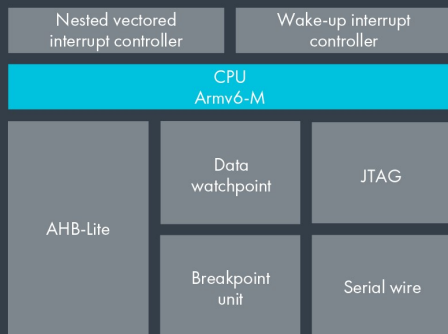
3- ARM CORTEX-M ARCHITECTURE

- History of ARM
- Cortex-M Features 
- Cortex-M Instruction Set Architecture
- Instruction Set
- Memory Access
 - Memory-mapped I/O
- Exception Handling

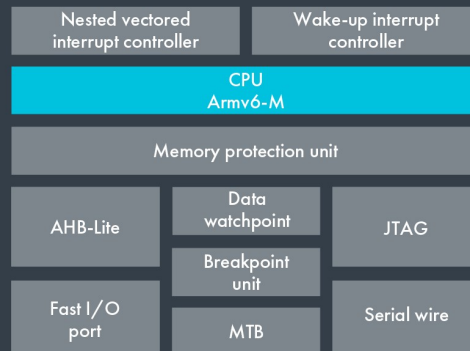
CORTEX-M FAMILY

INTRODUCING THE CORTEX-M FAMILY - 1/2

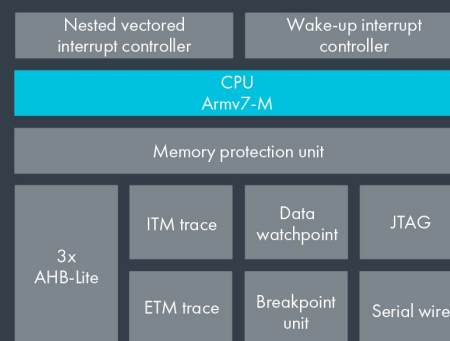
arm CORTEX®-M0



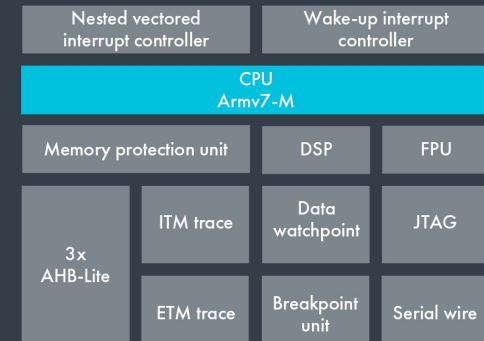
arm CORTEX®-M0+



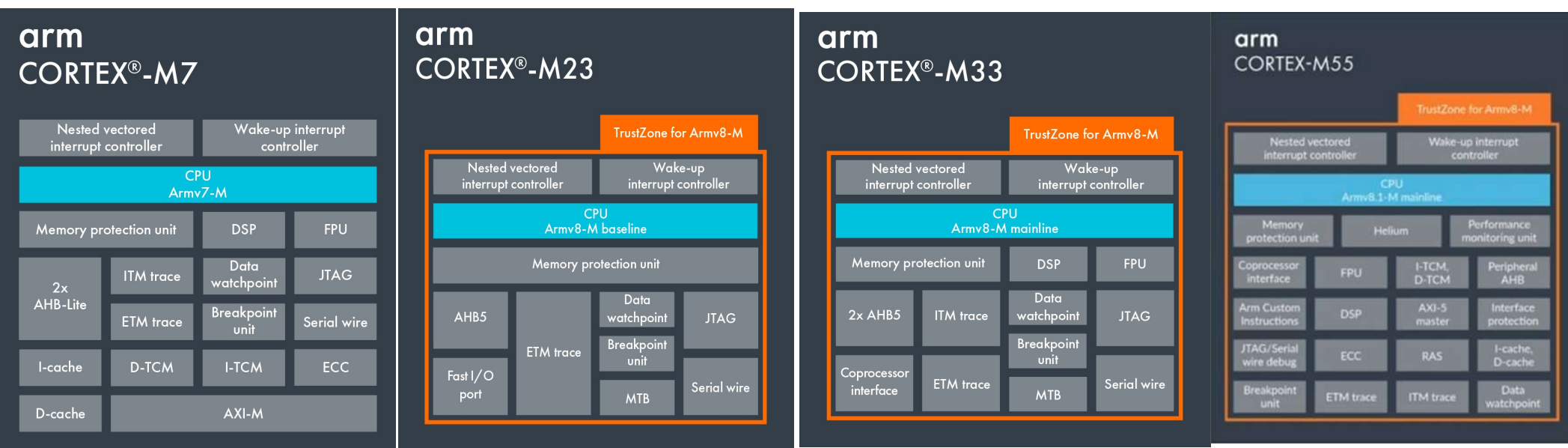
arm CORTEX®-M3



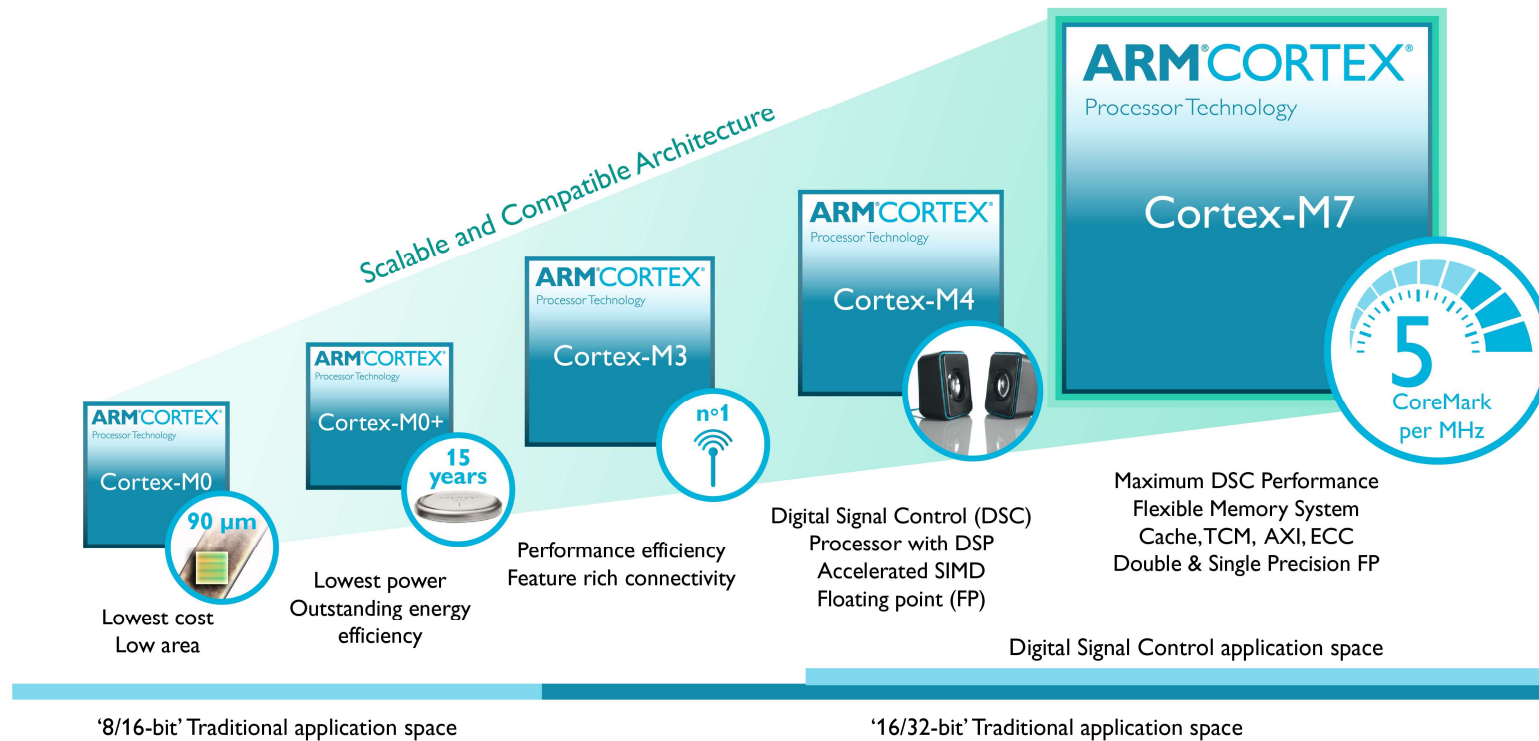
arm CORTEX®-M4



INTRODUCING THE CORTEX-M FAMILY - 2/2

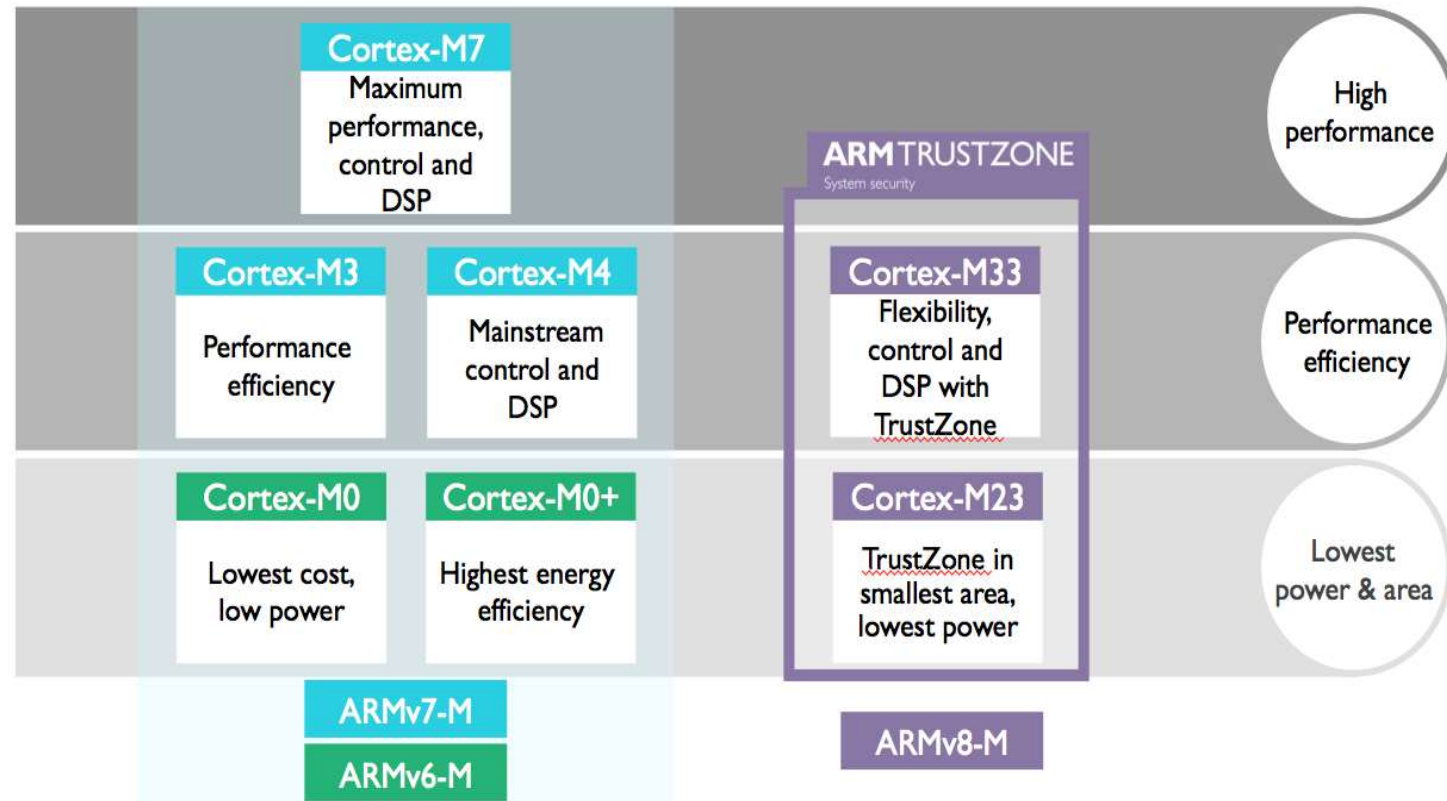


THE CORTEX-M FAMILY



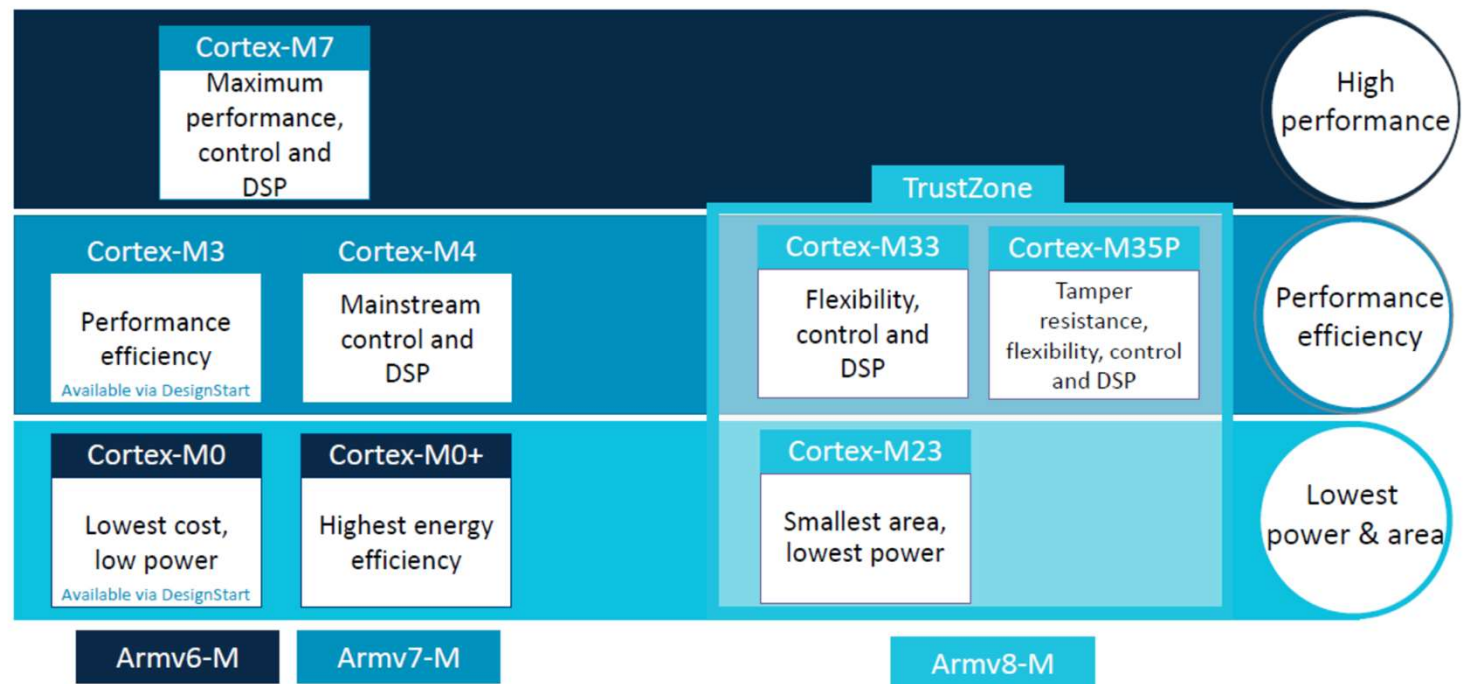
THE CORTEX-M FAMILY

In 2016 the Cortex-M33 and Cortex-M23 were added to the Cortex-M family.



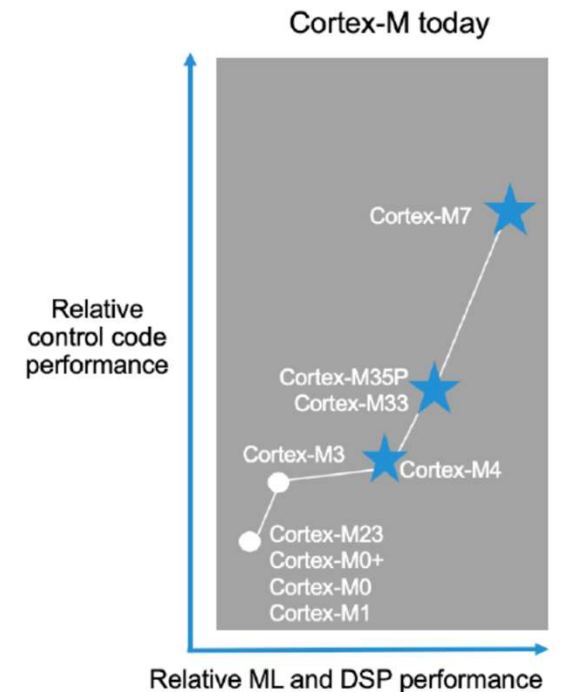
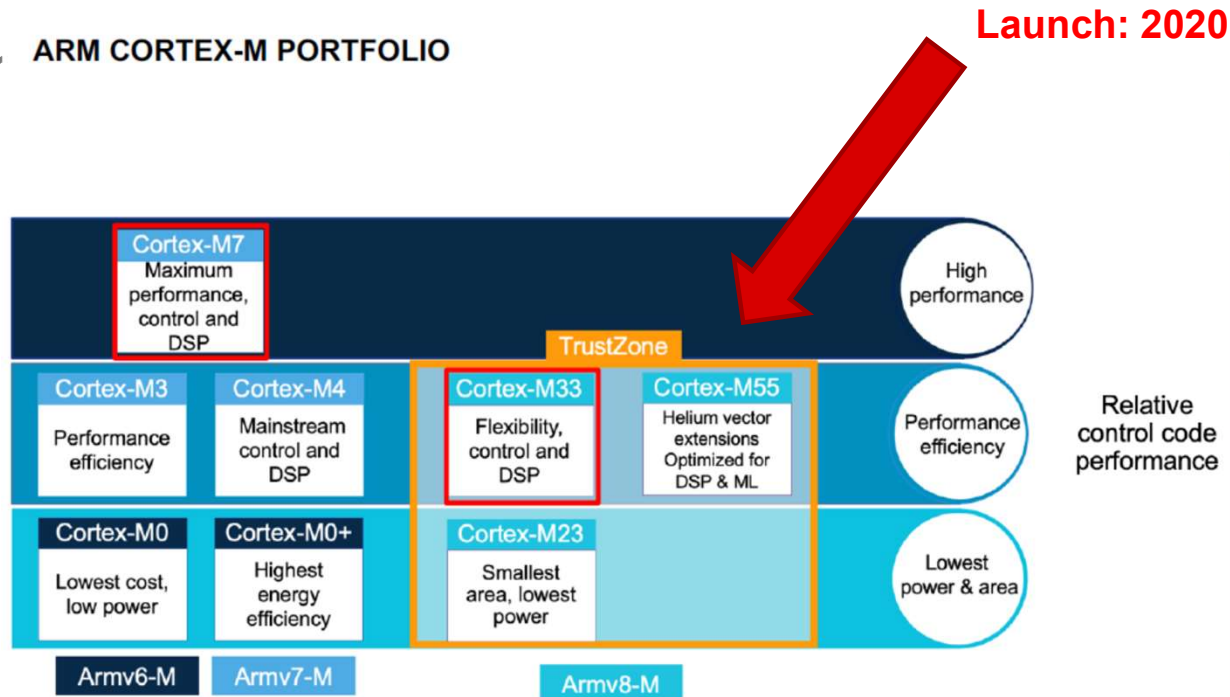
THE CORTEX-M FAMILY

And in 2018 the Cortex-M35P was added to the Cortex-M family.

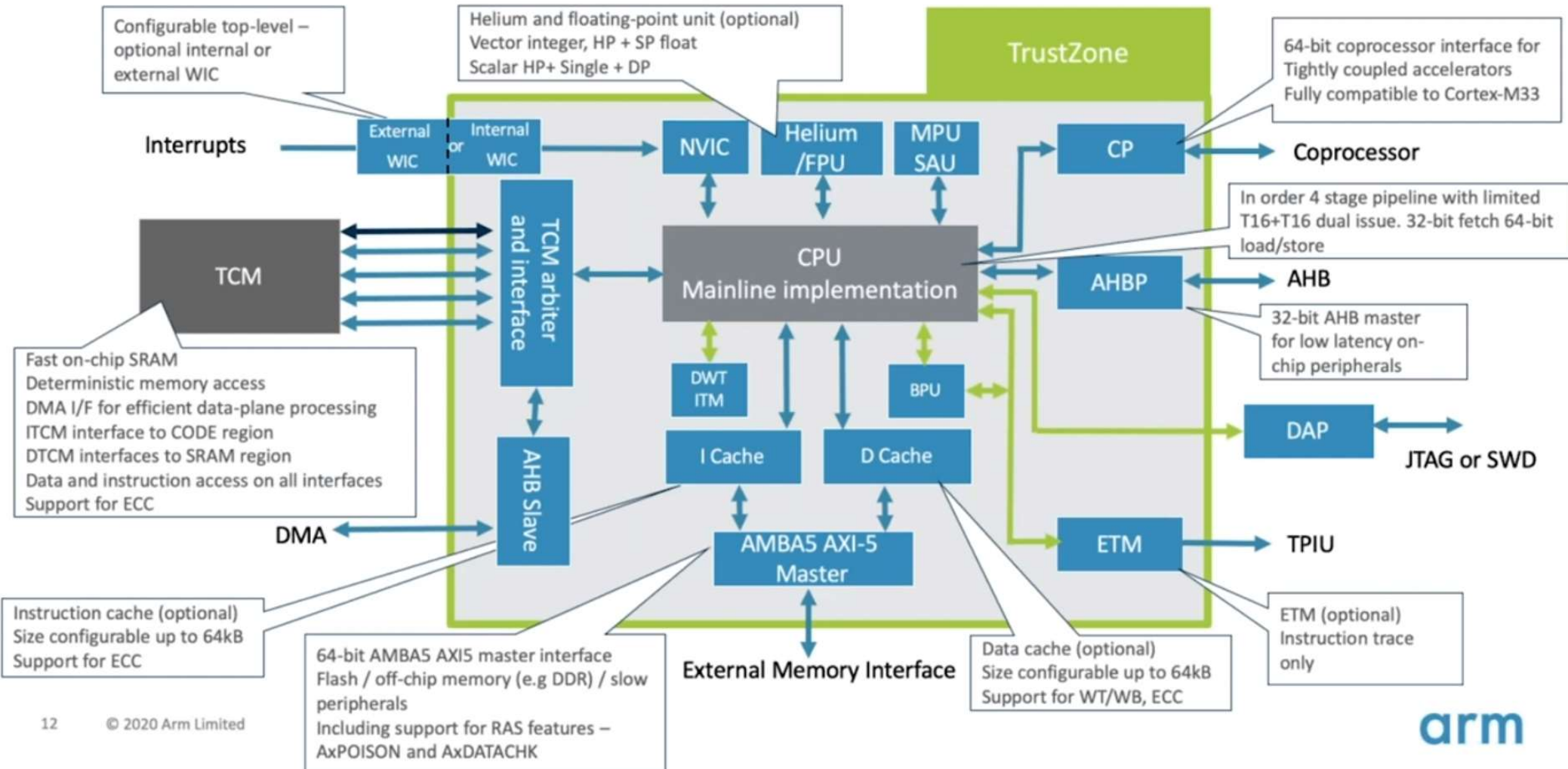


THE CORTEX-M FAMILY TODAY (2021)

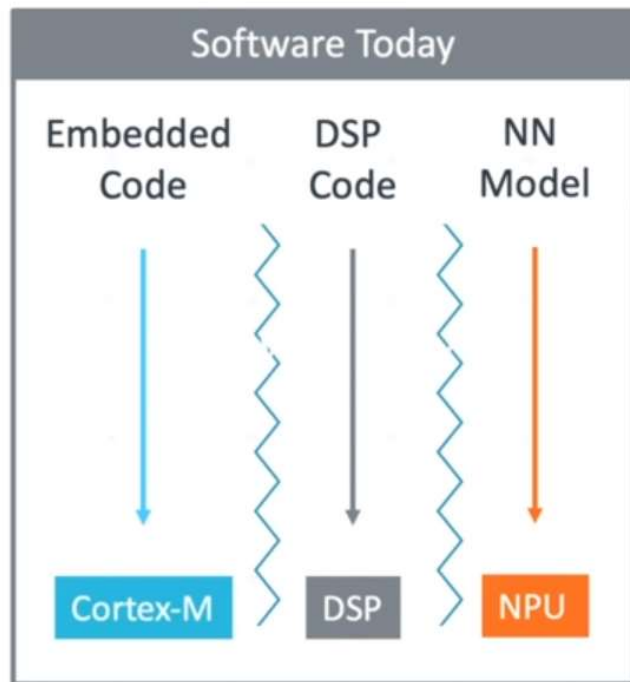
ARM CORTEX-M PORTFOLIO



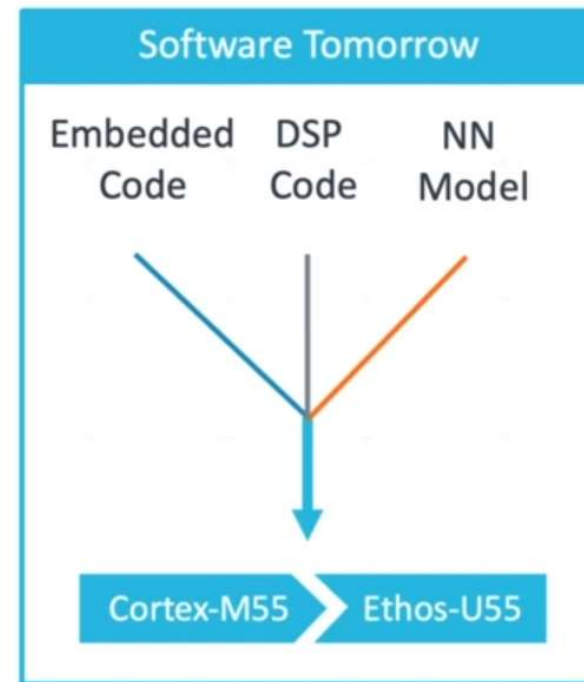
Block Diagram of the Cortex-M55 Processor



Unified Software Development: Fastest Path to Endpoint AI



- Multiple software development flows
- Harder to program and debug
- More complex, longer time to market



- Unified software development flow
- Works with common ML frameworks and existing tools
- More productivity, faster time to market

Cortex-M55: The Most AI-capable Cortex-M Processor

Cortex-M processor with enhanced DSP/ML compute capabilities

Helium vector processing

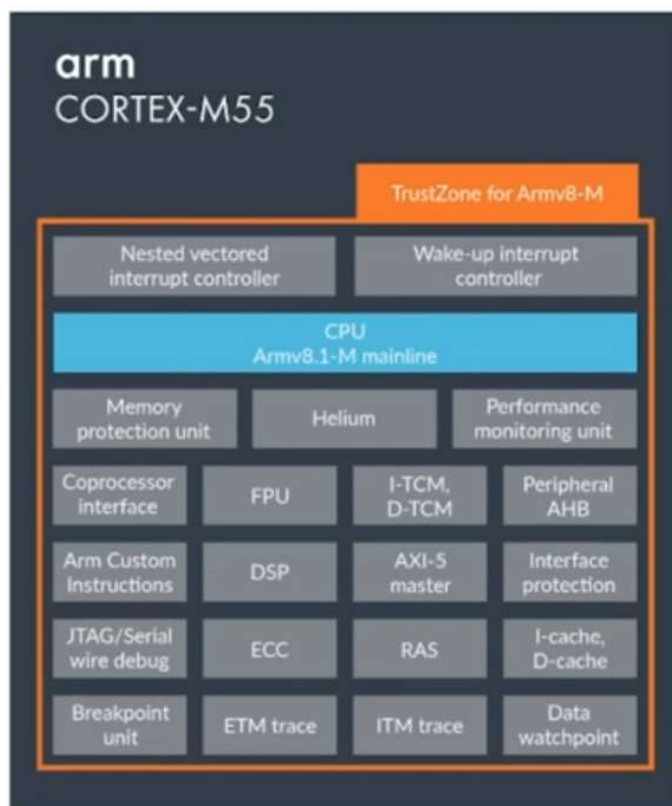
- > 150 new scalar and vector instructions
- Support for complex maths
- Low overhead loops

Vector processing support

- 2 x 32-bit MAC/cycle
- 4 x 16-bit MAC/cycle
- 8 x 8-bit MAC/cycle

Extended datatype support

- Half-precision float
- 8-bit integer
- Floating-point (half, full and double-precision)



Cortex-M ease of use

- Unified instruction set
- Single toolchain, simplified debug
- No need for separate DSP engine
- Cortex-M ecosystem

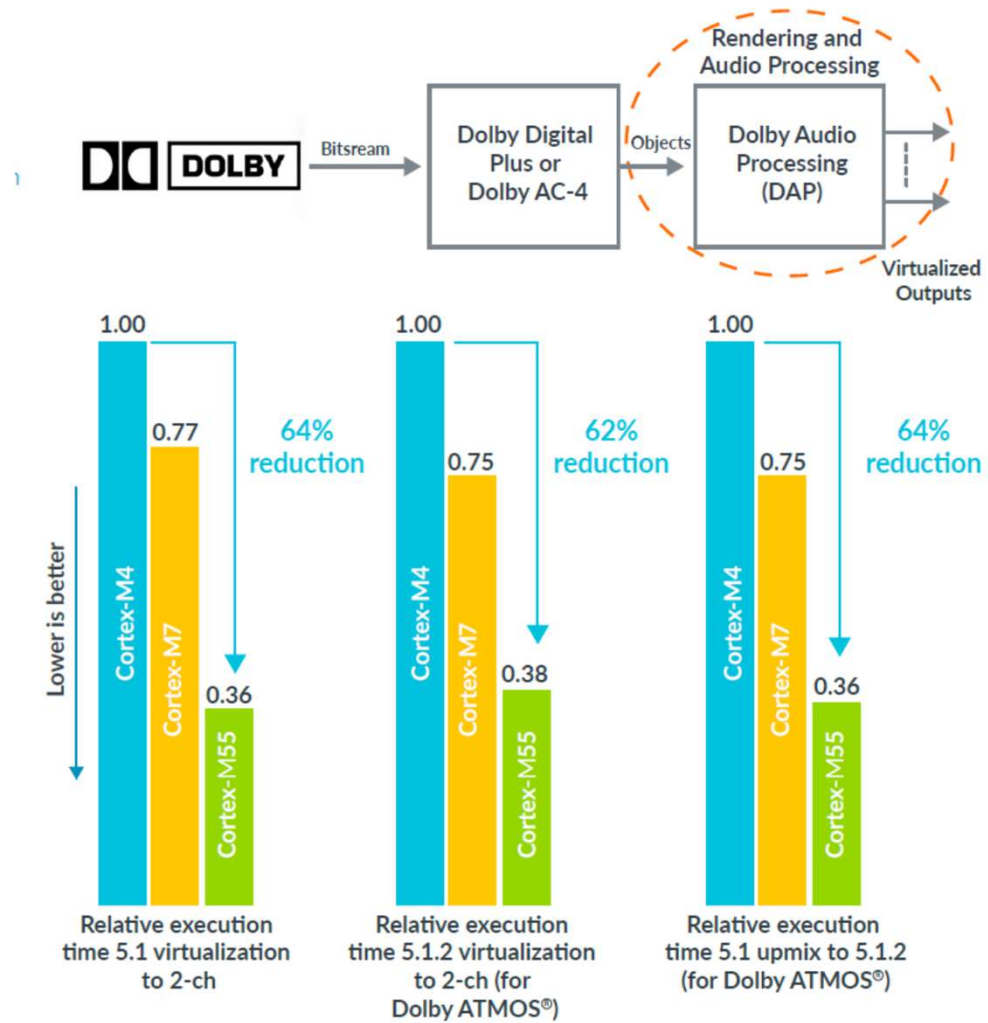
High performance system

- Memory system designed for DSP and ML applications
- Optional I/D caches
- Optional Tightly Coupled Memory

Security

- TrustZone for system wide security
- Protect software investments

CORTEX-M55 COMPARISON



Processor	Descriptions
Cortex-M0	A very small processor (starting from 12K gates) for low cost, ultra low power microcontrollers and deeply embedded applications
Cortex-M0+	The most energy-efficient processor for small embedded system. Similar size and programmer's model to the Cortex-M0 processor, but with additional features like single cycle I/O interface and vector table relocations
Cortex-M1	A small processor design optimized for FPGA designs and provides Tightly Coupled Memory (TCM) implementation using memory blocks on the FPGAs. Same instruction set as the Cortex-M0
Cortex-M3	A small but powerful embedded processor for low-power microcontrollers that has a rich instruction set to enable it to handle complex tasks quicker. It has a hardware divider and Multiply-Accumulate (MAC) instructions. In addition, it also has comprehensive debug and trace features to enable software developers to develop their applications quicker
Cortex-M4	It provides all the features on the Cortex-M3, with additional instructions target at Digital Signal Processing (DSP) tasks, such as Single Instruction Multiple Data (SIMD) and faster single cycle MAC operations. In addition, it also have an optional single precision floating point unit that support IEEE 754 floating point standard
Cortex-M7	High-performance processor for high-end microcontrollers and processing intensive applications. It has all the ISA features available in Cortex-M4, with additional support for double-precision floating point, as well as additional memory features like cache and Tightly Coupled Memory (TCM)
Cortex-M23	A small processor for ultra-low power and low cost designs, similar to the Cortex-M0+ processor, but with various enhancements in instruction set and system-level features. It also supports the TrustZone security extension.
Cortex-M33	A mainstream processor design, similar to previous Cortex-M3 and Cortex-M4 processors, but with much better flexibility in system design, better energy efficiency and higher performance. It also supports the TrustZone security extension.

THE CORTEX-M FAMILY - COMPARISON

Instruction Set Comparison (non-exhaustive list)

	Cortex-M0	Cortex-M0+	Cortex-M3	Cortex-M4	Cortex-M7
Single-cycle Multiply	Yes (32bit product)	Yes (32bit product)	Yes (32/64bit product)	Yes (32/64bit product)	Yes (32/64bit product)
Hardware divide	-	-	Yes	Yes	Yes
Advanced instructions	-	-	Yes	Yes	Yes
memory accesses	-	-	Yes	Yes	Yes
Table branch, compare & branch, conditional exec	-	-	Yes	Yes	Yes
Exclusive access	-	-	Yes	Yes	Yes
SIMD DSP	-	-	-	Yes	Yes
Saturation arithmetic	-	-	Adjust only	Full set	Full set
Float pointing (optional)	-	-	-	SP only	SP and DP

source: ARM

How to Choose Your ARM Cortex-M Processor - Tim Menasveta

THE CORTEX-M FAMILY - COMPARISON

System Features

	Cortex-M0	Cortex-M0+	Cortex-M3/M4	Cortex-M7	
Sleep modes	Yes	Yes	Yes	Yes	Low Power support
WIC, SRPG support	Yes	Yes	Yes	Yes	
OS support (SVC, PendSV)	Yes	Yes	Yes	Yes	OS support
MPU + Unprivileged	-	Optional (0 or 8 regions)	Optional (0 or 8 regions)	Optional (0, 8, or 16 regions)	Advanced OS Handling
Fault/Exception Handler	HardFault	HardFault	HardFault+3 configurable	HardFault+3 configurable	Fault handling
Fault Status Registers	-	-	Yes	Yes	
Bit Band	-	-	Yes (Optional)	Yes (Optional)	
Bus interface	AHB Lite	AHB Lite	AHB Lite, APB	AXI4, AHB Lite, APB, TCM	interface
L1 cache	-	-	-	Up to 64kB (I & D each)	
TCM (Tightly Coupled Memory)	-	-	-	Up to 1MB (I & D each)	

THE CORTEX-M FAMILY - COMPARISON

Cortex-M Performance Considerations

- CoreMark® and Dhrystone (per MHz)
 - Results depends on C compiler used
 - Might not be a good indication of the performance for your real world applications

Cortex-M7 for high processing requirements

- Up to 2x DSP performance of Cortex-M4
- Highest CoreMark

Cortex-M3/M4 can be faster because of

- Richer instruction set
- Harvard bus architecture
- Write buffer
- Speculative fetch of branch targets

Cortex-M0+ can be faster for simple I/O control tasks because of

- Shorter pipeline
- Single cycle I/O interface

Actual performance depends on

- System level design
- Memory wait states
- Clock configurations, etc

	Dhrystone (official)	Dhrystone (max opt)	CoreMark®
Cortex-M0	0.87	1.27	2.33
Cortex-M0+	0.95	1.36	2.46
Cortex-M3	1.25	1.89	3.32
Cortex-M4	1.25	1.95	3.40
Cortex-M7	2.14	3.23	5.01

* CoreMark data from ARM website and EEMBC.org website

3.2 - CORTEX-M FEATURES

Among the distinguishing features of the Cortex-M architecture are:

- 3-stage pipeline (except for Cortex-M7 / M55): Fetch, Decode, Execute,
- Harvard architecture (except for Cortex-M0 and M0+),
- Designed for power efficiency (includes an ultra low-power deep sleep),
- Thumb-2 instruction set, combining ARM performance and Thumb code density,
- Interrupt Controller (NVIC) is defined in the architecture; low latency vectored interrupt servicing,
- Interrupt servicing with tail-chaining and late arrival functionalities,
- Bit-banding to provide faster bit operations in memory and memory mapped I/O,
- MPU = Memory Protection Unit,
- Most instructions can be conditional.

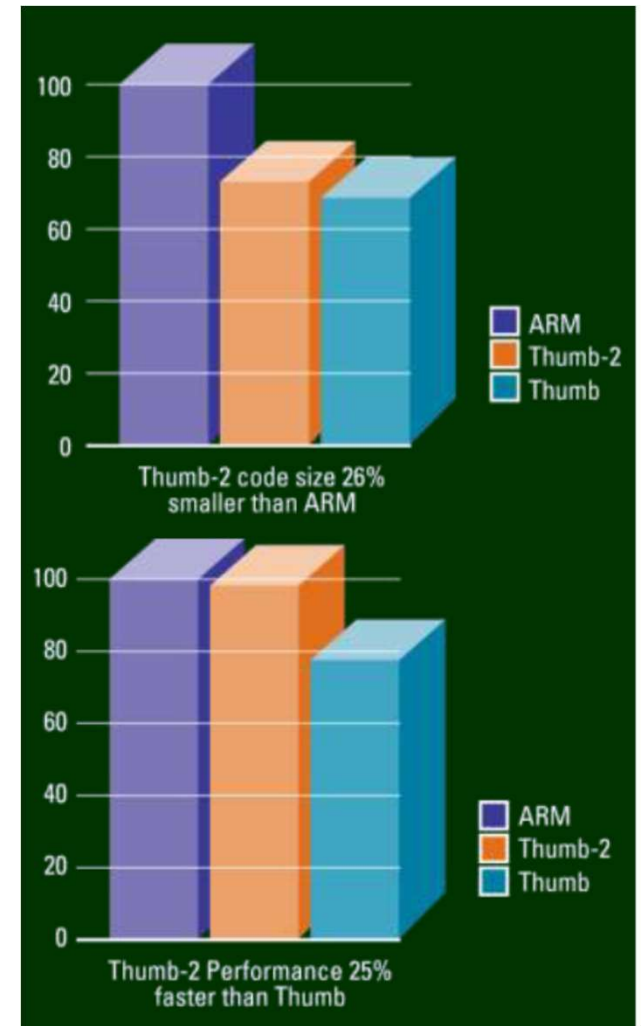
3.2 CORTEX-M FEATURES

The ARM7TDMI had two instructions sets:

- ARM - with 32-bit instructions, higher performance and lower code density,
- Thumb - with 16-bit instructions, lower performance and higher code density.

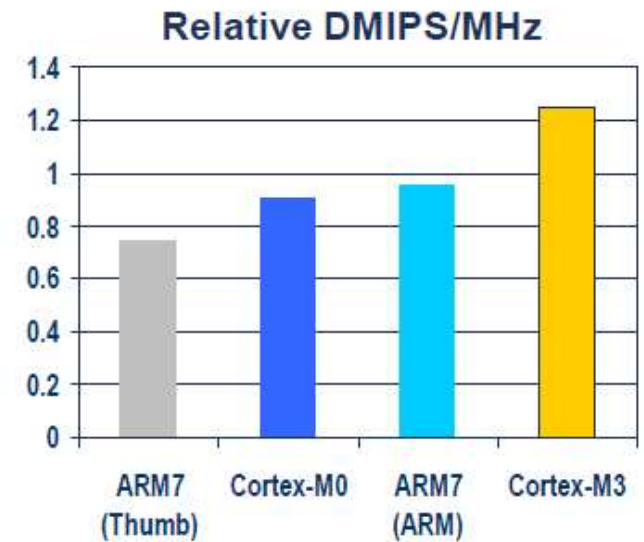
Cortex-M has a single instruction set: Thumb-2

It mixes 16-bit and 32-bit instructions. Its code density is similar to Thumb and its performance is similar to ARM.



3.2 CORTEX-M FEATURES

Comparison	Cortex-M0	Cortex-M3
DMIPS/MHz	0.9	1.25
Gate count	12k	43k
Number interrupts	1-32 + NMI	1-240 + NMI
Interrupt priorities	4	256
Breakpoints, Watchpoints	4/2, 2/1	8/4, 2/1
MPU, integrated trace option	No	Yes
Hardware Divide	No	Yes



source: ARM
ARM Cortex-M Processor Family

3.2 CORTEX-M FEATURES

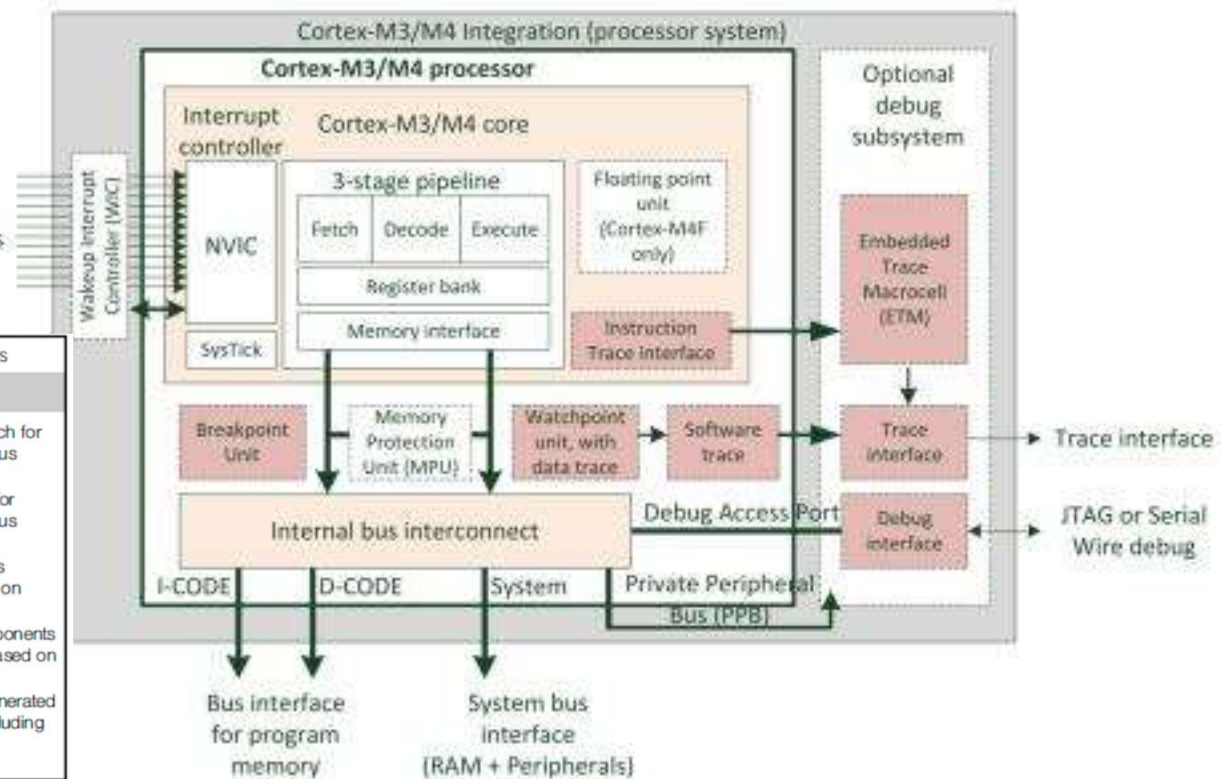
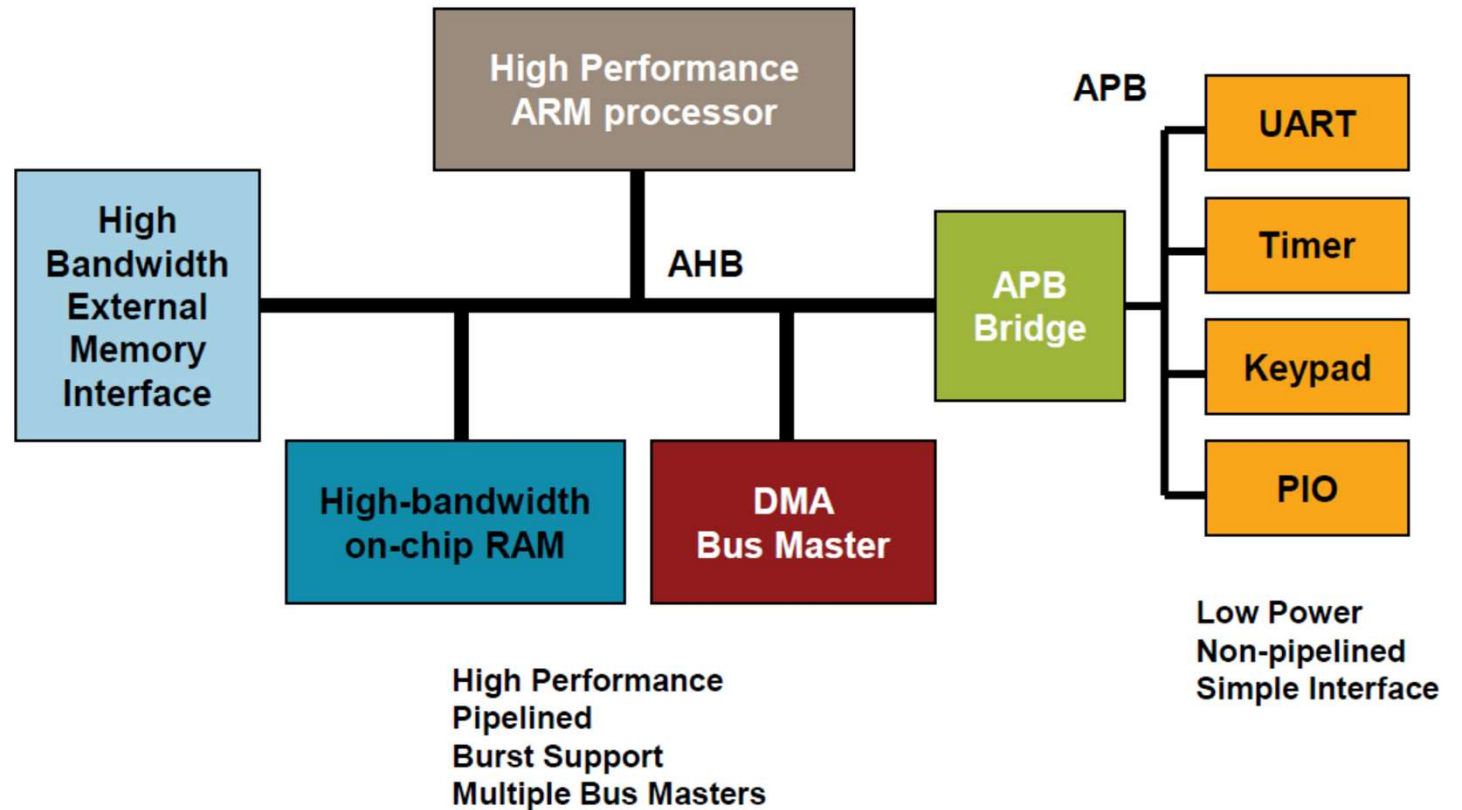


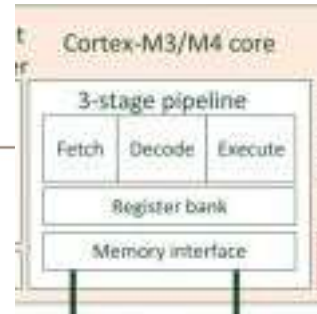
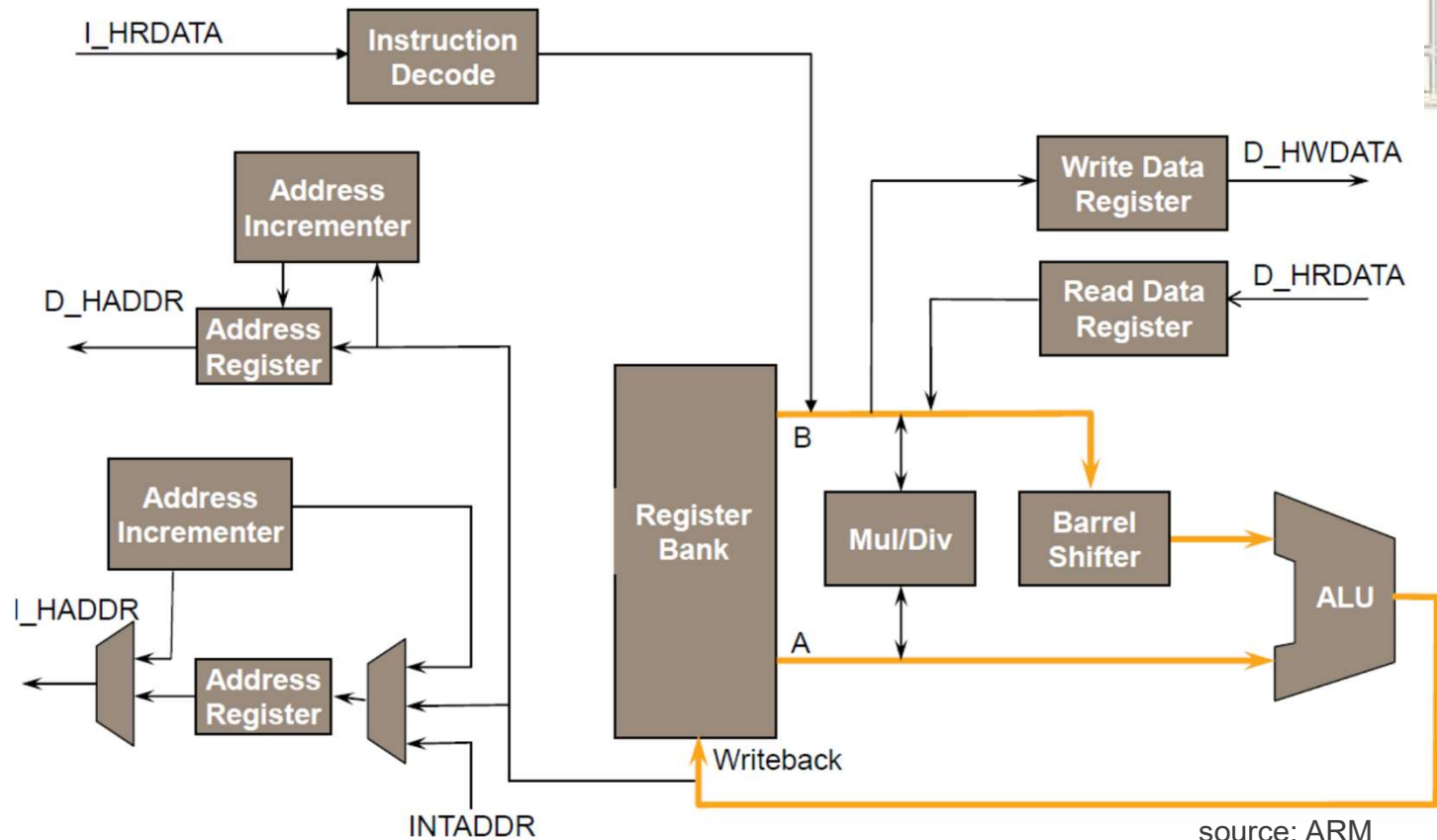
Table 3.2 Various Bus Interfaces on the Cortex-M3 and Cortex-M4 Processors

Bus Interface	Descriptions
I-CODE	Primarily for program memory: Instruction fetch and vector fetch for address 0x0 to 0x1FFFFFFF. Based on AMBA 3.0 AHB Lite bus protocol.
D-CODE	Primarily for program memory: Data and debugger accesses for address 0x0 to 0x1FFFFFFF. Based on AMBA 3.0 AHB Lite bus protocol.
System	Primarily for RAM and peripherals: Any accesses from address 0x20000000 to 0xFFFFFFFF (apart from PPB regions). Based on AMBA 3.0 AHB Lite bus protocol.
PPB	External Private Peripheral Bus (PPB): For private debug components on system level from address 0xE0040000 to 0xE00FFFFF. Based on AMBA 3.0 APB protocol.
DAP	Debug Access Port (DAP) interface: For debugger accesses generated from the debug interface module to any memory locations including system memory and debug components. Based on the ARM CoreSight™ debug architecture.

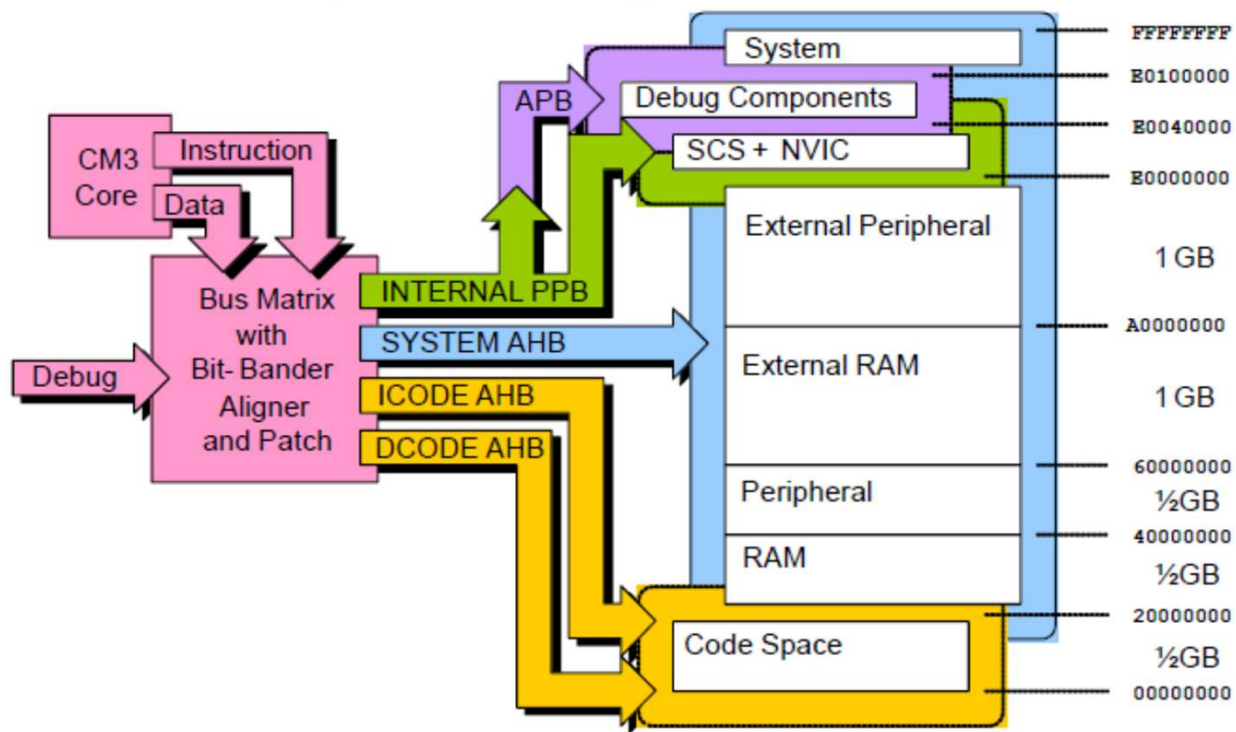
3.2 CORTEX-M FEATURES



Cortex-M3 Datapath



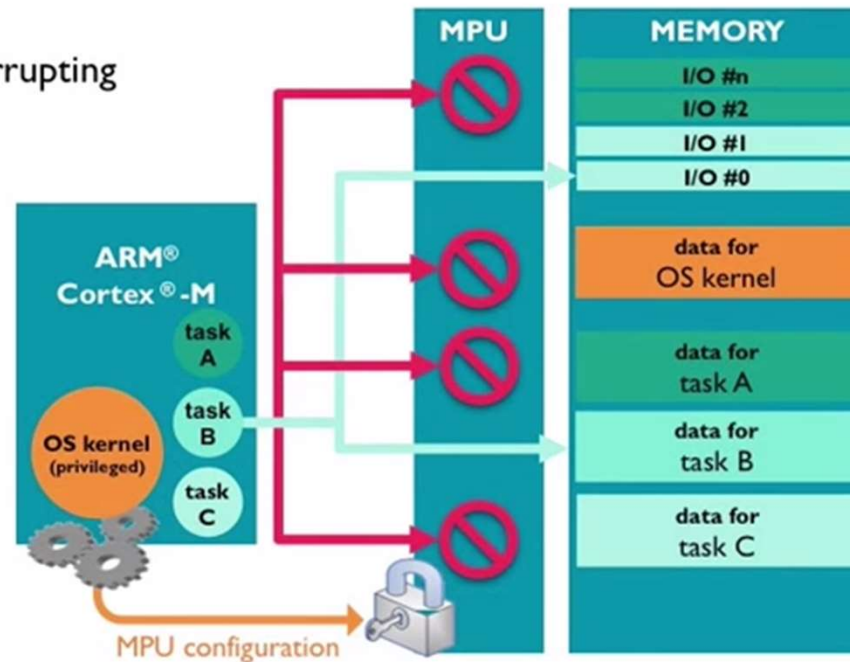
source: ARM
The ARM Architecture - Joe Bungle



THE CORTEX-M FAMILY - FEATURES

Memory Protection Unit (MPU)

- Prevents application task from corrupting OS or other task data
 - Improves system reliability
- Configurable regions
 - Address
 - Size
 - Memory attributes
 - Access permissions
- Optional in all processors (except Cortex-M0)



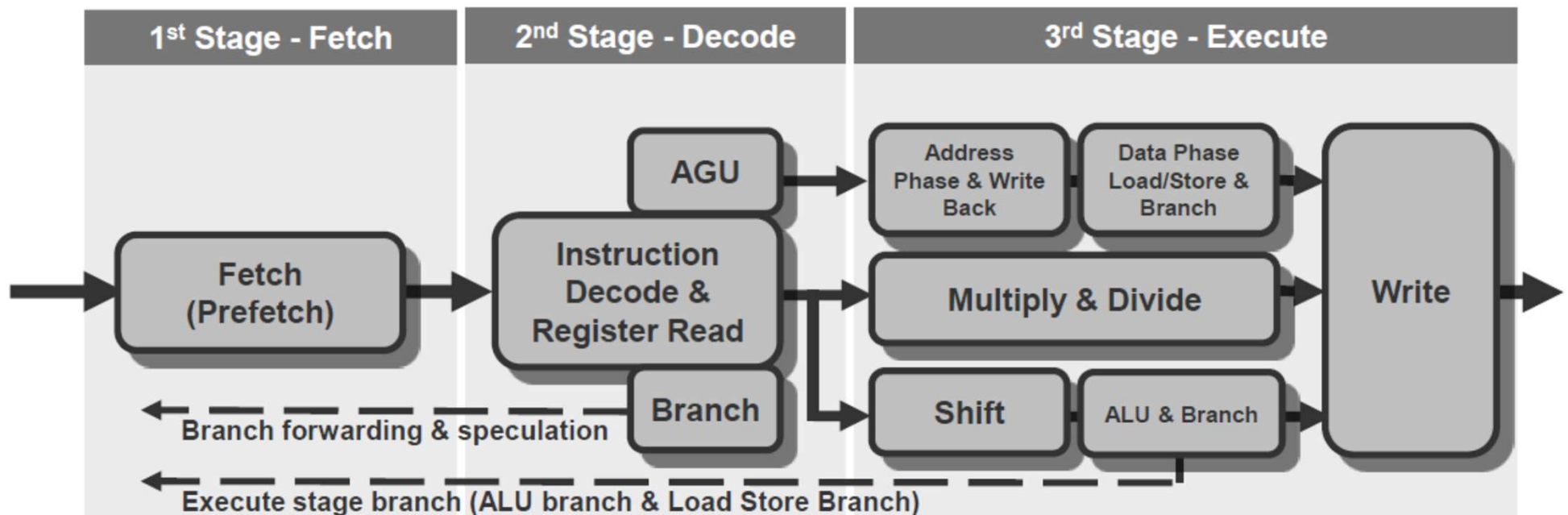
PIPELINE

Instructions are executed in one of these sequences:

1. FETCH
DECODE
EXECUTION (Logic./ARIT.)
2. FETCH
DECODE
EXECUTION (Data address calculation)
DATA MEMORY ACCESS

3.2 CORTEX-M FEATURES

Cortex-M4 Pipeline



source: ARM
The ARM Architecture - Joe Bungo

3.2 CORTEX-M FEATURES

Cortex-M4 Pipeline - example for optimal execution

Cycle		1	2	3	4	5	6	7	8	9
Operation										
ADD		F	D	E						
SUB			F	D	E					
ORR				F	D	E				
AND					F	D	E			
ORR						F	D	E		
EOR							F	D	E	

F - Fetch D - Decode E - Execute

3.2 CORTEX-M FEATURES

Cortex-M4 Pipeline - pipeline flush due to indirect branch (no forwarding)

Cycle		1	2	3	4	5	6	7	8	9
Address	Operation									
0x8000	BX r5	F	D	E						
0x8002	SUB		F	D						
0x8004	ORR			F						
0x8FEC	AND			F	D	E				
0x8FEE	ORR				F	D	E			
0x8FF0	EOR				F	D	E			

F - Fetch D - Decode E - Execute

3.2 CORTEX-M FEATURES

Cortex-M4 Pipeline

- Harvard architecture allows for concurrent access to code and data memory.

Cycle		1	2	3	4	5	6	7	8	9
Operation										
ADD	F	D	E							
SUB		F	D	E						
STR			F	D	Ea	Ed				
STR				F	D	Ea	Ed			
ORR					F	D	E			
EOR						F	D	E		

F - Fetch **D** - Decode **E** - Execute **S** - Stall
Ea - Execute / STR address phase **Ed** - STR data phase

3.2 CORTEX-M FEATURES

Cortex-M4 Pipeline

Figure 2. Pipeline stages in the Cortex-M3 and Cortex-M4 processors

